

ALI RAZA KHALID

Machine Learning & Deep Learning Researcher

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SUMMARY

BS Artificial Intelligence student (Semester 3, CGPA 3.91/4.0) focused on computer vision and deep learning. I have trained CNNs and U-Net models for image tasks, built neural networks from scratch, and optimized a model to run fast on CPUs. Actively reading research papers and testing new architectures and methods for deep learning.

SKILLS

Programming: Python, C++, JavaScript, TypeScript, SQL

Deep Learning: PyTorch, Scikit-learn, NumPy, Pandas, CNNs, Transformers, RNN/LSTM/GRU, Model Training and Evaluation

Computer Vision: OpenCV, Image Classification, Image Segmentation (U-Net), Preprocessing and Augmentation

Edge Deployment: C++ with AVX2, INT8 vector operations, CPU inference, Model Optimization

Tools: Git/GitHub, Linux, Docker, Jupyter, Google Colab, Kaggle

Foundations: Linear Algebra, Probability, Calculus for Machine Learning

RELEVANT PROJECTS

Image Segmentation with U-Net

Python, PyTorch

- Built and trained a U-Net model that predicts a mask for each object in an image.
- Added image augmentation to predict masks more accurately. Reached a validation Dice score of 0.85 and IoU of 0.76.
- Code: github.com/Razamindset/image-segmentation-with-unet

CNN Study on CIFAR

Python, PyTorch

- Improved a CNN on CIFAR from 75% to about 87% accuracy over several rounds of testing.
- Tested dropout, BatchNorm with GELU, a deeper 3-layer conv stack, and a learning rate scheduler. Fixed a data augmentation leak between train and eval.

Indus Dragon: Chess Engine with Neural Network

C++, PyTorch, AVX2

- Trained a custom MLP on 25 million self-play positions and deployed it for fast CPU inference using AVX2 and INT8 vector operations.
- Reached 3M+ nodes per second and an ELO of 2200+. This shows I can make models small and fast for edge devices. [Code](#)

Neural Networks and Sequence Models from Scratch

Python, NumPy

- Built MLPs, CNNs, RNNs, LSTMs, GRUs and a full Transformer without ML libraries, including Adam, SGD and RMSprop optimizers.
- Studied how architecture choices affect performance on long sequences. Also built a GPT-1 style Transformer.

Super Brain: WhatsApp AI Assistant

Python, PostgreSQL, pgvector

- Built and deployed an AI chatbot with a custom retrieval pipeline and REST API on Oracle Cloud.

OPEN SOURCE AND EXPERIENCE

- Contributor, Chesskit** and **Infinitunes & Co-founder, ScreenSnipper** <https://screensnipper.app>

EDUCATION

BS Artificial Intelligence, CGPA 3.91/4.0

Expected 2028

PAF-IAST, Haripur, Pakistan. Coursework: Machine Learning, Deep Learning, Cloud Computing, Algorithms and Data Structures.